

What is Object Detection in Computer Vision?

Object Detection is a computer vision task that identifies and locates multiple objects within an image or video. Unlike image classification, which assigns a label to an entire image, object detection predicts both the object category and its location using bounding boxes.

- Enables real-time analysis of images and video streams.
- Forms the foundation of applications such as autonomous driving, surveillance, and robotics.

Types

1. **Image Classification:** Assigns a single label to the entire image based on its content. It determines what is present in the image but does not indicate the object's location.
2. **Object Localization:** Identifies an object and determines its position within the image by drawing a bounding box around it.
3. **Object Detection:** Combines image classification and localization to identify multiple objects in an image, assign labels to them, and provide their locations using bounding boxes.

Working

1. **Input Image:** The process begins with an input image or video frame.
2. **Pre-processing:** The image is resized, normalized, or transformed into a suitable format for the model.
3. **Feature Extraction:** Important visual features are extracted to identify object patterns.
4. **Classification:** Detected regions are classified into predefined object categories.
5. **Localization:** Bounding boxes are generated to determine the location of each object.
6. **Non-Maximum Suppression (NMS):** Overlapping bounding boxes are filtered to retain the most accurate detections.
7. **Output:** The final image is displayed with labeled bounding boxes around the detected objects.